

Let's do together

(1) Show that the following argument is valid. If Mohan is a lawyer, then he is ambitious. If Mohan is an early riser, then he does not like idlies. If Mohan is ambitious, then he is an early riser.

Then if Mohan is a lawyer, then he does not like idlies.

(2) Show that the hypothesis "If you send me an e-mail message, then I will finish writing the program. If you do not send me an email message, then I will go to sleep early. If I go to sleep early, then I will wake up feeling refreshed" leads to the conclusion "If I do not finish writing the program, then I will wake up feeling refreshed."

(3) Show that the following argument is valid. If today is Tuesday, I have a test in Mathematics or Economics. If my Economics Professor is sick, I will not have a test in Economics. Today is Tuesday and my Economics Professor is sick. Therefore, I have a test in Mathematics.

Why Predicates?

Date

“Propositional logics are not adequate to express the meaning of all statements in natural language”

e.g:- Suppose we know that “Every computer connected to the university network is functioning properly”.

Now, there ~~are~~ ^{that} no rules of propositional logic allow us to conclude the truth of the statement. “MATH3 is functioning properly.”

where MATH3 is one of the computers connected to the university network.

And so we introduce a more powerful type of logic called predicate logic.

Predicates

statements involving variables, such as

“ $x > 3$ ”, “ $x = y + 3$ ”, “ $x + y = z$ ”, “computer x is under attack by an intruder”, “computer x is functioning properly”, are often found in mathematical assertions, in computer programs, and in system specifications.

These statements are neither true nor false when the value of variables are not specified.

Statement “ x is greater than 3” has two parts.

↓	↓
Variable	Predicate
(x)	(P)

So the statement will be denoted by $P(x)$. *Spiral*

Date

e.g:- (1) Let $P(x)$ denote the statement " $x > 3$ ", what are the truth values of $P(4)$ and $P(2)$?

(2) Let $Q(x, y)$ denote the statement " $x = y + 3$ ". What are the truth values of $Q(1, 2)$ and $Q(3, 0)$?

(3) Let $A(c, n)$ denote the statement "Computer c is connected to network n ". where c is a variable representing a computer and n is a variable representing a network. Suppose that MATH1 is connected to network CAMPUS2 but not to network CAMPUS1. What are the truth values of $A(\text{MATH1}, \text{CAMPUS1})$ and $A(\text{MATH1}, \text{CAMPUS2})$?

(4) If $x > 0$ then $x = x + 1$

$P(x)$

If $P(x)$ is true for this value of x then $x = x + 1$ and if $P(x)$ is false then the assignment statement is not executed.

Universe of Discourse

As discussed above, Predicates become propositions once every variable is bound by assigning it a value from the universe of Discourse U or quantifying it.

OR

The collection of values that a variable x can take is called x 's Universe of Discourse.

n-place predicate

Predicate logic generalizes the grammatical notion of a predicate to also include propositional functions of any number of arguments.

e.g:- Let R be the three-variable predicate

$$R(x, y, z) : x + y = z.$$

Find the truth values of $R(2, -1, 5)$, $R(3, 4, 7)$.